

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO5 - ASSESSMENT TOOL 1 – Pseudocode Design

Course Code:	DSA0404	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO5 – Pseudocode Design	Assessment:	Assessment Tool 1– Pseudocode Design
Weightage:	30% of CIA	Total Marks:	100
CO5 Statement:	To understand the concept of supervised and unsupervised methods in machine learning		
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Section / Batch:	2024	Date:	24-08-2026

Code of Conduct

I _ T.Chandana _ (192472059)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _ T.Chandana _

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

11.Design pseudocode to train a linear regression model, calculate predicted values and residuals, and determine the Mean Squared Error (MSE) and R^2 score.

Answer:

Problem Statement

Design pseudocode to train a linear regression model, calculate predicted values and residuals, and determine the Mean Squared Error (MSE) and R^2 score.

Algorithm

1. Start.
2. Load the dataset containing input features and target values.
3. Separate the input features (X) and target variable (Y).
4. Split the dataset into training and testing data.
5. Create a Linear Regression model.
6. Train the model using the training data.
7. Predict the target values for the test data.
8. Calculate residuals using actual values minus predicted values.
9. Calculate the Mean Squared Error (MSE).
10. Calculate the R^2 score.
11. Display the predicted values, residuals, MSE, and R^2 score.
12. Stop.

Pseudocode

BEGIN

LOAD dataset

```
SELECT input features X
SELECT target variable Y
SPLIT X and Y into training data and testing data
CREATE Linear Regression model
TRAIN model using X_train and Y_train
PREDICT Y_pred using X_test
CALCULATE residuals
residuals = Y_test - Y_pred
CALCULATE MSE
MSE = MEAN((Y_test - Y_pred)^2)
CALCULATE R2 score
R2 = 1 - [SUM((Y_test - Y_pred)^2) / SUM((Y_test - MEAN(Y_test))^2)]
DISPLAY Y_pred
DISPLAY residuals
DISPLAY MSE
DISPLAY R2
END
```

Output

Predicted Values:

[72.5, 81.2, 65.8, 90.4, 76.3]

Residuals:

[2.5, -1.2, 4.2, -0.4, 1.7]

Mean Squared Error (MSE):

4.12

R² Score:

0.92

12.Design pseudocode for logistic regression to predict whether a student will pass or fail based on study hours and attendance percentage.

Answer:

Problem Statement

Design pseudocode for logistic regression to predict whether a student will pass or fail based on study hours and attendance percentage.

Algorithm

1. Start.
2. Load the student dataset.
3. Select study hours and attendance percentage as input features.
4. Select pass/fail status as the target variable.
5. Split the dataset into training and testing data.
6. Create a Logistic Regression model.
7. Train the model using the training data.
8. Give study hours and attendance percentage as input.
9. Predict whether the student will pass or fail.
10. Display the prediction.
11. Stop.

Pseudocode

BEGIN

LOAD student dataset

SELECT Study_Hours and Attendance as input features X

SELECT Pass_Fail as target variable Y

SPLIT X and Y into training and testing data

CREATE Logistic Regression model

TRAIN model using X_train and Y_train

```
INPUT study_hours
INPUT attendance
CREATE student_data = [study_hours, attendance]
PREDICT class using the trained model
IF predicted class = 1 THEN
    DISPLAY "PASS"
ELSE
    DISPLAY "FAIL"
END IF
END
```

Output

Input:

Study Hours = 6

Attendance = 85%

Predicted Result:

PASS

13.Design pseudocode that calculates the sigmoid probability for logistic regression and converts the probability into a binary class using a threshold of 0.5.

Answer:

Problem Statement

Design pseudocode that calculates the sigmoid probability for logistic regression and converts the probability into a binary class using a threshold of 0.5.

Algorithm

1. Start.
2. Obtain the weighted input value z .
3. Calculate the sigmoid probability.

4. Compare the probability with the threshold 0.5.
5. If probability is greater than or equal to 0.5, assign class 1.
6. Otherwise, assign class 0.
7. Display the probability and predicted class.
8. Stop.

Pseudocode

```
BEGIN
INPUT z
CALCULATE sigmoid probability
probability =  $1 / (1 + \text{EXP}(-z))$ 
SET threshold = 0.5
IF probability >= threshold THEN
    predicted_class = 1
ELSE
    predicted_class = 0
END IF
DISPLAY probability
DISPLAY predicted_class
END
```

Output

Example:

Input $z = 1.5$

Sigmoid Probability = 0.8176

Threshold = 0.5

Predicted Class = 1

14.Design pseudocode to evaluate a logistic regression classifier using the confusion matrix, accuracy, precision, recall, and F1-score.

Answer:

Problem Statement

Design pseudocode to evaluate a logistic regression classifier using the confusion matrix, accuracy, precision, recall, and F1-score.

Algorithm

1. Start.
2. Obtain the actual class values.
3. Obtain the predicted class values from the logistic regression model.
4. Construct the confusion matrix.
5. Identify True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN).
6. Calculate accuracy.
7. Calculate precision.
8. Calculate recall.
9. Calculate F1-score.
10. Display all evaluation measures.
11. Stop.

Pseudocode

BEGIN

INPUT actual values Y_actual

INPUT predicted values Y_pred

CALCULATE confusion matrix

FIND:

TP = True Positive

TN = True Negative

FP = False Positive

FN = False Negative

CALCULATE accuracy

Accuracy = $(TP + TN) / (TP + TN + FP + FN)$

CALCULATE precision

Precision = $TP / (TP + FP)$

CALCULATE recall

Recall = $TP / (TP + FN)$

CALCULATE F1-score

F1 = $2 \times (\text{Precision} \times \text{Recall}) /$
 $(\text{Precision} + \text{Recall})$

DISPLAY confusion matrix

DISPLAY accuracy

DISPLAY precision

DISPLAY recall

DISPLAY F1-score

END

Output

Confusion Matrix:

	Predicted	
	0	1
Actual 0	45	5
1	3	47

TP = 47

TN = 45

FP = 5

FN = 3

Accuracy = 92%

Precision = 90.38%

Recall = 94.00%

F1-Score = 92.16%

15.Unsupervised Design pseudocode for an unsupervised learning process that accepts Learning an unlabelled dataset, preprocesses the features, discovers groups, and produces cluster assignments.

Answer:

Problem Statement

Design pseudocode for an unsupervised learning process that accepts an unlabelled dataset, preprocesses the features, discovers groups, and produces cluster assignments.

Algorithm

1. Start.
2. Load the unlabelled dataset.
3. Select the relevant features.
4. Check for missing values.
5. Handle missing values if present.
6. Standardize the features.
7. Select a clustering algorithm such as K-Means.
8. Choose the number of clusters.
9. Train the clustering algorithm on the processed data.

10. Generate cluster assignments for each data point.

11. Display the cluster assignments.

12. Stop.

Pseudocode

BEGIN

LOAD unlabelled dataset

SELECT relevant features

CHECK for missing values

IF missing values exist THEN

 HANDLE missing values

END IF

STANDARDIZE the features

SET number of clusters K

CREATE K-Means clustering model

FIT model using preprocessed data

GENERATE cluster assignments

DISPLAY cluster assignments

END

Output

Example:

Number of Clusters = 3

Cluster Assignments:

Data Point 1 → Cluster 0

Data Point 2 → Cluster 1

Data Point 3 → Cluster 0

Data Point 4 → Cluster 2

Data Point 5 → Cluster 1

Data Point 6 → Cluster 2

Data Point 7 → Cluster 0

Data Point 8 → Cluster 1